Mobile Applications Development

CS 442

F2025

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Mobile Computing Concepts

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Entities we come across …

- Device
  - Mobile
  - Tablet
- OS
  - Windows
  - Symbian*
  - Android
  - webOS
  - iOS
  - RIM
  - KaiOS
- Framework/ API/ IDE
  - Android Studio
  - Flutter
  - ReactNative
  - Xcode (iOS)
- Prog. Lang.
  - Java
  - ActiveScript
  - Swift
  - Kotlin
  - C++
  - Dart

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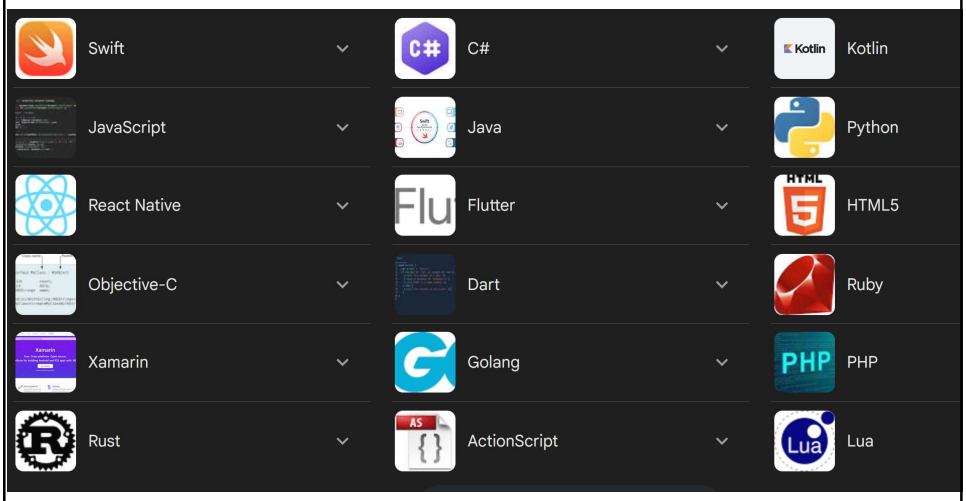
Languages for Mobile App Development



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API/Framework

React Native	Xamarin	Flutter
Kotlin	Ionic	Java
C#	JavaScript	Python
Swift	Apache Cordova	NativeScript
Ruby	jQuery Mobile	Angular
Objective-C	CSS	Dart
PHP	Build mobile and web applica...	Solar2D
Framework7	Go	Adobe PhoneGap

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Operating system Software / Mobile Phone

From sources across the web

Android GNU General Public License	Windows Phone Proprietary software	iOS Proprietary software
Microsoft Windows Proprietary software	webOS GNU General Public License	Windows 10 Mobile
Linux GNU General Public License	BlackBerry 10 Proprietary software	B2G OS Mozilla Public License
ChromeOS Proprietary software	watchOS Proprietary software	Tizen BSD licenses
KaiOS GNU General Public License	AliOS Proprietary software	macOS Proprietary software
LineageOS Apache License	Android Ice Cream Sandwich	Windows 10 Shareware
Windows Phone 7	LuneOS GNU General Public License	Windows 8 Proprietary software
CopperheadOS	GrapheneOS MIT License	Android One
iPadOS 14 Proprietary software	Windows 11 Proprietary software	

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History - Wireless Networks

- 1947 : AT&T Bell Labs : Conceived the idea of cellular phones
- 1958 : Germany: Wireless Network commissioned : A-Netz : analog
- Evolved to B-Netz
- 1968 : AT & T proposed a cellular network
- 1973 : Motorola : First mobile handset
- 1983 : First American commercial analog cellular service : AMPS (Advanced Mobile Phone Service)
- 1979 : Japan had its first commercial analog cellular service
- Europe : Every country developed their own systems ; incompatible to each other. The incompatibility made problems for equipment manufacturers.
- 1982 : Study group GSM created in Europe – Groupe Speciale Mobile – Standard for pan-European
- 1991 – Commercial service of GSM ('digital') started. Because of its popularity outside Europe, GSM : Global System for Mobile communications

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Mobility

- Physical
  - Movement of matter
- Virtual
  - Movement of bits and bytes

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Mobile Computing – Different Contexts

- Mobile Computing : computing environment is mobile and moves along with the user.
  - Offline and Real time : user is able to use all his remote data and services online
- Anywhere, anytime information
- Virtual Home Environment : An environment in foreign network
- Nomadic :Computing environment moves along with the mobile user
- Pervasive : Environment that can be made available in any environment
- Ubiquitous : Machines that fit the human environment instead of forcing humans to enter theirs .
- Global Service Portability: Any service of any environment available globally
- Wearable computers : Body Area Network

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Mobile Computing Functions

- User Mobility
- Network Mobility
- Bearer Mobility
- Device Mobility
- Session Mobility
- Service Mobility
- Host Mobility

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Mobility

User Mobility: User should be able to move from one physical location to another location and use the same service. The service could be in the home network or a remote network. Example could be a user moves from London to New York and uses Internet to access the corporate application the same way the user uses in the home office.

Network Mobility: User should be able to move from one network to another network and use the same service. Example could be a user moves from Hong Kong to New Delhi and uses the same GSM phone to access the corporate application through WAP (Wireless Application Protocol). In home network he uses this service over GPRS (General Packet Radio Service) whereas in Delhi he accesses it over the GSM network.

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Mobility

Bearer Mobility: User should be able to move from one bearer to another and use the same service. Example could be a user was using a service through WAP bearer in his home network in Bangalore. He moves to Coimbatore, where WAP is not supported, he switch over to voice or SMS (Short Message Service) bearer to access the same application.

Device Mobility: User should be able to move from one device to another and use the same service. Example could be sales representatives using their desktop computer in home office. During the day while they are on the street they would like to use their Palmtop to access the application.

Session Mobility: A user session should be able to move from one user-agent environment to another. Example could be a user was using his service through a CDMA (Code Division Multiple Access) 1X network. The user entered into the basement to park the car and got disconnected from his CDMA network. User goes to home office and starts using the desktop. The unfinished session in the CDMA device moves from the mobile device to the desktop computer.

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Mobility

Service Mobility: User should be able to move from one service to another. Example could be a user is writing a mail. To complete the mail user needs to refer to some other information. In a desktop PC, user simply opens another service (browser) and moves between them using the task bar. User should be able to switch amongst services in small footprint wireless devices like in the desktop.

Host Mobility: The user device can be either a client or server. When it is a server or host, some of the complexities change. In case of host mobility the mobility of IP needs to be taken care of.

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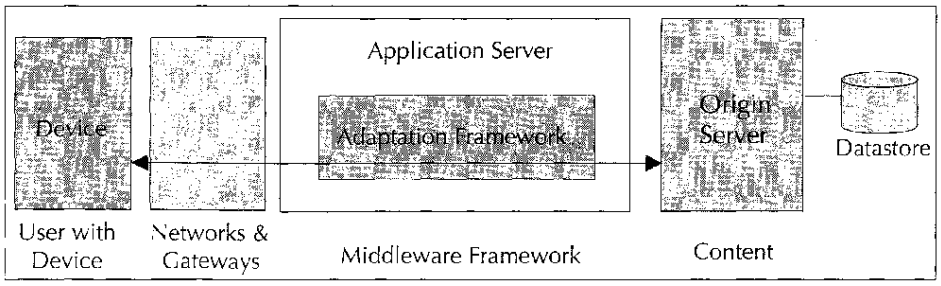
Mobile Computing Functions – Segments

- User with device,
- Network & Gateway
- Middleware
- Content



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Middleware

- Communication
  - Application communicate with different nodes and services through different communication middleware
- Transaction processing
  - Session maintenance
- Behaviour management
  - Client Device and application
- Communication Gateway
  - Dissimilar protocols handling

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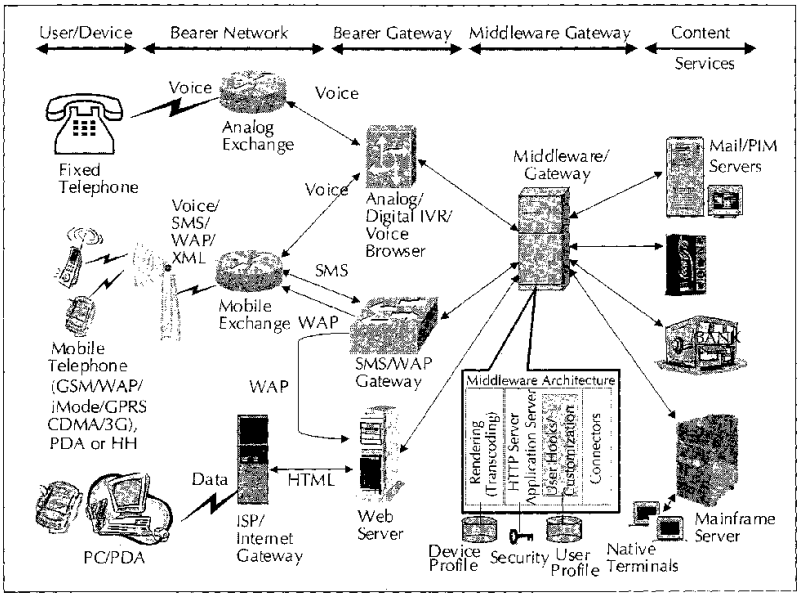
Mobile Computing environment



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Applications and Services (Contents)

Data and information through mobile computing services.

Categories for Mobile applications

- Personal
- Perishable
- Transaction Oriented
- Location Specific
- Corporate
- Entertainment

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Categories for mobile applications

Personal–belongs to the user (wallet, life-tool, medical records, diary).

Perishable–time sensitive and relevance passes quickly (general news, breaking news, weather, sports, business news, stock quotes).

Transaction Oriented–transactions need to be closed (bank transactions, utility bill payment, mobile shopping).

Location Specific–information related to current geographical location (street direction map, restaurant guide).

Corporate–corporate business information (mail, ERP, inventory, directory, business alerts, reminders).

Entertainment–applications for fun, entertainment.

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Developing Mobile computing applications

- New Mobile Application
- Making Legacy application Mobile

- Special Challenge
  - Security issues

- Standards
  - **Standards are documented agreements containing technical specifications or other precise criteria to be used consistently as rules, guidelines or definitions of characteristics**

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Mobile Computing Architecture

- Three-tier Network-centric mobile computing architecture

  - Layer 1: User Interface or Presentation Tier
  - Layer 2: Process Management or Application Tier
  - Layer 3: Database Management or Data Tier

  Design Patterns!

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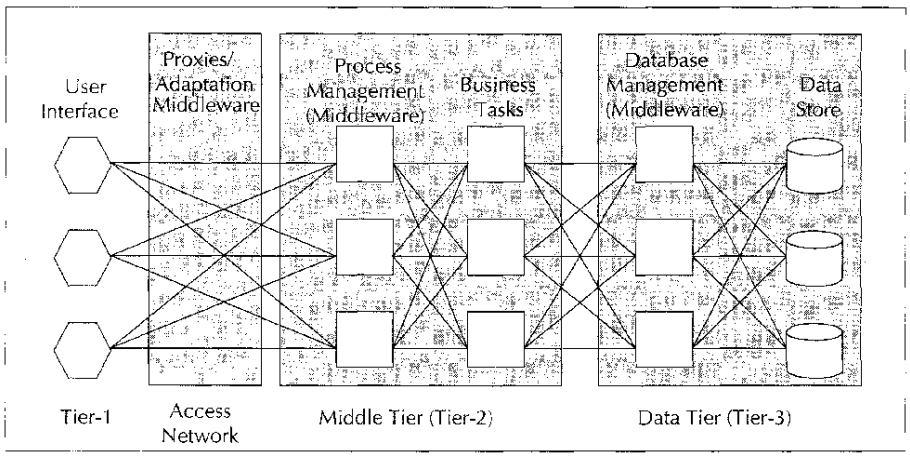
Three-tier Architecture



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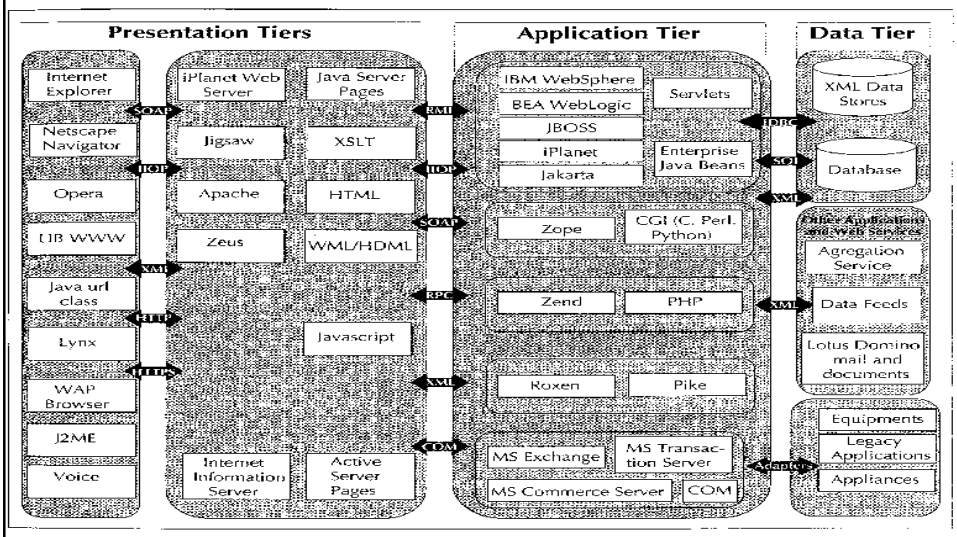
Mobile Computing Architecture (Old)



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Presentation Tier

This is the user facing system in the first tier. This is the layer of agent applications and systems. These applications run on the client device and offer all the user interfaces. This tier is responsible for presenting the information to the end user. Humans generally use visual and audio means to receive information from machines (with some exceptions like vibrator in mobile phones). Humans also use keyboard (laptop computers, cell phones), pen (tablet PC, palmtops), touch screen (kiosks), or voice (telephone) to feed the data to the system. In the case of the visual, the presentation of information will be through a screen. Therefore, the visual presentation will relate to rendering on a screen. 'Presentation Tier' includes web browsers (like Mozila, lynx, Internet Explorer and Netscape Navigator), WAP browsers and customized client programs. A mobile computing agent needs to be context-aware and device-independent.

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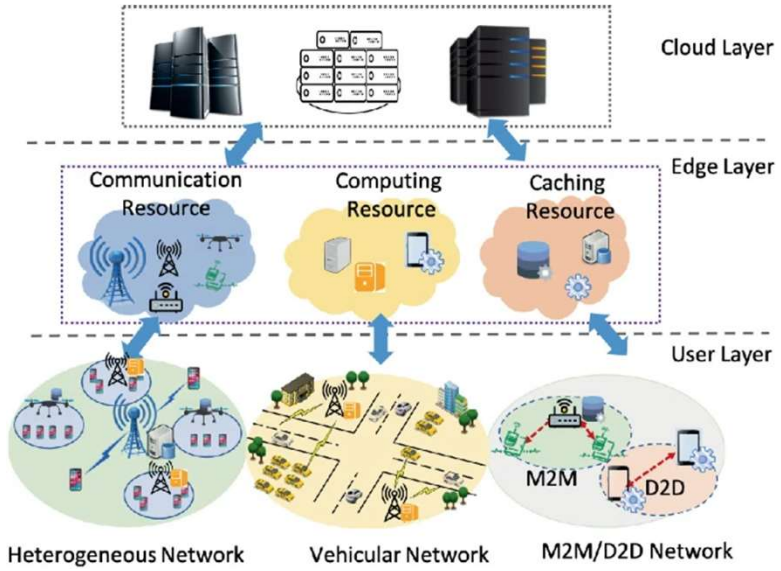
Mobile Edge Cloud



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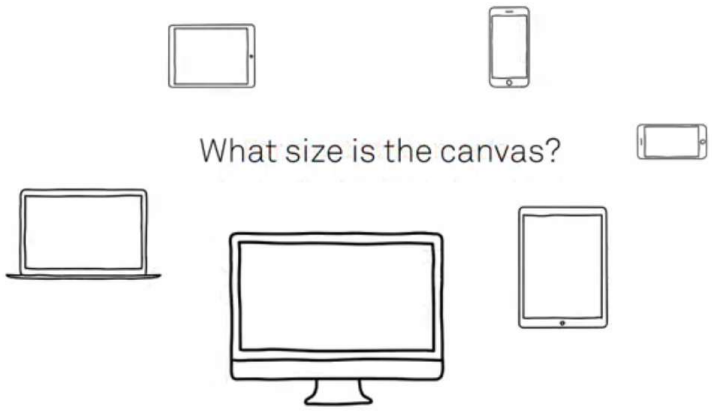
Websites/Apps do not have one specific size!



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Application Tier

The application tier or middle tier is the 'engine' of a ubiquitous application. It performs the business logic of processing user input, obtaining data, and making decisions. In certain cases, this layer will do the transcoding of data for appropriate rendering in the Presentation Tier. The Application Tier may include technology like CGI's, Java, JSP, .NET services, PHP or ColdFusion, deployed in products like Apache, WebSphere, WebLogic, iPlanet, Pramati, JBOSS or ZEND. The application tier is presentation and database-independent.

1. Message-Oriented Middleware
2. Transaction Processing Middleware
3. Database Middleware
4. Communication Middleware
5. Distributed Object and Components
6. Transcoding Middleware.

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Message-Oriented

Message-oriented Middleware is a middleware framework that loosely connects different applications through asynchronous exchange of messages. A MOM works over a networked environment without having to know what platform or processor the other application is resident on. The message can contain formatted data, requests for action, or unsolicited response. The MOM system provides a message queue between any two interoperating applications. If the destination process is out of service or busy, the message is held in a temporary storage location until it can be processed. MOM is generally asynchronous, peer-to-peer, and works in publish/subscribe fashion. In publish/subscriber model one or many objects subscribe to an event. As the event occurs, it will be published by the asynchronous loosely coupled object. The MOM will notify the subscribers about this event. However, most implementations of MOM support synchronous (request/response) message passing as well. MOM is most appropriate for event-driven applications. When an event occurs, the publisher application hands off to the messaging middleware application the responsibility of notifying subscribers that the event has happened. In a netcentric environment, MOM can work as the integration platform for different applications. Example of MOM are Message Queue from IBM known as MQ Series. The equivalent from Java is JMS (Java Message Service).

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Transaction Processing

Transaction Processing Middleware provides tools and an environment for developing transaction-based distributed applications. An ideal TP system will be able to input data into the system at the point of information source and the output of the system is delivered at the point of information sink. In an ideal TP system, the device for input and output can potentially be different (Figure 2.3). Also, the output can be an unsolicited message for a device. TP is used in data management, network access, security systems, delivery order processing, airline reservations, customer service, etc. to name a few. TP systems are generally capable of providing services to thousands of clients in a distributed client/server environment. CICS (Customer Information Control System) is one of the early. TP application systems on IBM mainframe computers.

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Communication

Communication Middleware is used to connect one application to another application through some communication middleware, like connecting one application to another application through telnet. These types of middleware are quite useful in the telecommunication world. There are many elements in the core telecommunication network where the user interface is through telnet. A mediation server automates the telnet protocol to communicate to these nodes in the network. Another example could be to integrate legacy applications through proprietary communication protocols like TN5250 or TN3270.

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Distributed

An example of distributed objects and components is CORBA (Common Object Request Broker Architecture). CORBA is an open distributed object computing infrastructure being standardized by the Object Management Group (http://www.omg.org). CORBA simplifies many common network programming tasks used in a netcentric application environment. These are object registration, object location, and activation; request demultiplexing; framing and error-handling; parameter marshalling and demarshalling; and

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Transcoding

Transcoding Middleware is used to transcode one format of data to another format to suit the need of the client. For example, if we want to access a web site through a mobile phone supporting WAP, we need to transcode the HTML page to WML page so that the mobile phone can access it. Another example could be accessing a map from a PDA. The same map, which can be shown in a computer, needs to be reduced in size to fit the PDA screen. Technically transcoding is used for content adaptation to fit the need of the device. Content adaptation is also required to meet the network bandwidth needs. For example, some frames in a video clip need to be dropped for a low bandwidth network. Content adaptation used to be done through proprietary protocols. To allow interoperability, IETF has accepted the Internet Content Adaptation Protocol (ICAP). ICAP is now standardized and described in RFC3507.

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Data Tier

The Data Tier is used to store data needed by the application and acts as a repository for both temporary and permanent data. The data could be stored in any form of datastore or database. These can range from sophisticated relational database, legacy hierarchical database, to even simple text files. The data can also be stored in XML format for interoperability with other system and datasources. A legacy application can also be considered as a data source or a document through a communication middleware.

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Database Middleware

We have discussed that for a mobile computing environment, the business logic should be independent of the device capability. Likewise, though not essential, it is advised that business logic should be independent of the database. Database independence helps the maintenance of the system better. Database middleware allows the business logic to be independent and transparent of the database technology and the database vendor. Database middleware runs between the application program and the database. These are sometimes called database connectors as well. Example of such middleware will be ODBC, JDBC, etc. Using these middleware, the application will be able to access data from any data source. Data sources can be text files, flat file, spreadsheets, or a network, relational, indexed, hierarchical, XML database, object database, etc. from vendors like Oracle, SQL, Sybase, etc.

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Design Considerations for Mobile Computing

- Mobile environment needs to be
  - Context Independent
  - Context Sensitive
- Context information is the information related to the surrounding environment of an actor in that environment.
- Context : All the information that help determine the state of an object (or actor).
- Object can be a
  - Person
  - Device
  - Place
  - Physical object
  - Computational object
  - Surrounding Environment
  - Any entity being tracked by the system

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- Context Data is captured so that decisions can be made about how to adapt content or behavior to suit this context
- Mobility implies that attributes associated with devices and users will change constantly. These changes mean that content and behavior of applications should be adapted to suit the current situation.
- Examples of adaptation:
  - Content with context awareness
  - Content switch on context
  - Content transcoding on context

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1. **Content with context awareness** Build each application with context awareness. There are different services for different client context (devices). For example a bank decides to offer mobile banking application through Internet, PDA and mobile phone using WAP. These services are different and are http://www.mybank.com/inet.html, http://www.mybank.com/palm.html and http://www.mybank.com/wap.wml, respectively. The service http://www.mybank.com/inet.html assumes that the user will use computers to access this service. Therefore it is safe to offer big pages with text box, drop down menu. Also, it is fine to add a few animated pictures for the new product the bank is launching. We know that http://www.mybank.com/palm.html is a service for a PalmOS PDA. As the display size is small, we design the screen to be compact for the PDA and do not offer the same product animation. For the WAP service at http://www.mybank.com/wap.wml, we do a completely different user interface; we make all drop down options available through the option button in the mobile phone and remove all the graphics and animations.

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2. **Content switch on context** Another way is to provide intelligence for the adaptation of content within the service. This adaptation happens transparent to the client. In this case the service is the same for Internet, PDA and WAP. All access the bank's service through http://www.mybank.com/. A intelligent piece of code identifies the agent to decide what type of device or context it is. This intelligent code does the adaptation at run time based upon the agent in hand. The simplest way to do this is to look at the User-Agent value at the HTTP header and decide whether to route the request to http://mybank.com/inet.html or http://www.mybank.com/palm.html or http://www.mybank.com/wap.wml.

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3. **Content transcoding on context.** Another way is to provide an underlying middleware platform that performs the adaptation of the content based on the context and behavior of the device. This adaptation happens transparent to the client and the application. The middleware platform is intelligent enough to identify the context either from the http parameters or additional customized parameters. In this case the service may be in html or XML, the middleware platform transcode the code from html (or XML) to html, and wml on the fly. It can also do the trasncoding based on policy so that the html generated for a computer is different from a PDA.

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Client Context Manager

When we humans interact with other persons, we always make use of the implicit situational information of the surrounding environment. We interpret the context of the current situation and react appropriately. For example, we can go close to a lion in a zoo, but definitely not in the wild. Or, a person discussing some confidential matter with another person observes the gestures and voice tone of the other person and reacts in an appropriate manner or changes the subject if someone shows up suddenly. When we use content through a PC within the four walls of the organization, we do not have any problem. A majority of the applications can safely assume that the context is the enterprise LAN. It can be assumed that the environment is secured; it can also be assumed that the user will be using the systems in a particular fashion using the browser standardized by the company. These applications are developed keeping the large screen (for mainly PC) and browsers in mind.

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A mobile computing application, on the other hand, needs to operate in dynamic conditions. This is due to various device characteristics and network conditions. This demands a reactive platform that can make decisions about how to respond to changes to device capability, user preferences, enterprise policy, network policy and many other environmental factors. Context can be used as the basis by which an adaptation manager or algorithm decides to modify content or application behavior. We therefore need a Client Context Manager to gather and maintain information pertaining to the client device, user, network and the environment surrounding each mobile device. All these information will be provided by a set of *Awareness Modules*.

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Awareness Modules

Awareness modules are sensors

of various kinds. These sensors can be hardware sensors or software sensors or a combination of these. A hardware sensor can be used to identify the precise location of a user; whereas, a software sensor can be used to determine the type of the user agent. These awareness modules can be in the device, network, or even in the middleware. We use the term middleware in a very generic context. A middleware can be a functional module in the content server, a proxy or an independent system. For example, an awareness module in the device will provide information about its capabilities. Another example could be a location manager that tracks the location and orientation of the mobile device.

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Information seen as context

1. **Identity:** The device will be in a position to communicate its identity without any ambiguity.
2. **Spatial information:** Information related to the surrounding space. This relates to location, orientation, speed, elevation and acceleration.
3. **Temporal information:** Information related to time. This will be time of the day, date, timezone and season of the year.
4. **Environmental information:** This is related to the environmental surroundings. This will include temperature, air quality, moisture, wind speed, natural light or noise level. This also includes information related to the network and network capabilities.
5. **Social situation:** Information related to the social environment. This will include who you are with, and people that are nearby; whether the user is in a meeting or in a party.

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6. **Resources that are nearby:** This will relate to the other accessible resources in the nearby surroundings like accessible devices, hosts or other information sinks.
7. **Availability of resources:** This will relate to information about the device in use. This will include battery power, processing power, persistence store, display, capabilities related to I/O (Input/Output) and bandwidth.
8. **Physiological measurements:** This relates to the physiological state of the user. This includes information like blood pressure, heart rate, respiration rate, muscle activity and tone of voice.
9. **Activity:** This relates to the activity state of the user. This includes information like talking, reading, walking and running.
10. **Schedules and agendas:** This relates to the schedules and agendas of the user.

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Context Aware System

A system is context-aware if it can extract, interpret and use context-related information to adapt its functionality to the current context. The challenge for such systems lies in the complexity of capturing, representing, filtering and interpreting contextual data. To capture context information generally some sensors are required. This context information needs to be represented in a machine-understandable format, so that applications can use this information. In addition to being able to obtain the context-information, applications must include some 'intelligence' to process the information and deduce the meaning. These requirements lead us to three aspects of context management:

1. **Context sensing:** The way in which context data is obtained.
2. **Context representation:** The way in which context information is stored and transported.
3. **Context interpretation:** The way in which meaning is obtained from the context representation.

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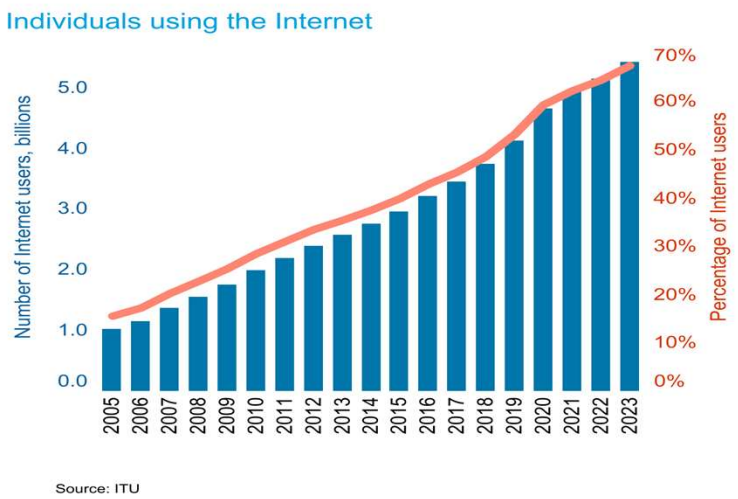
ITU

- https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx
- More than three quarter of world population owns a mobile phone



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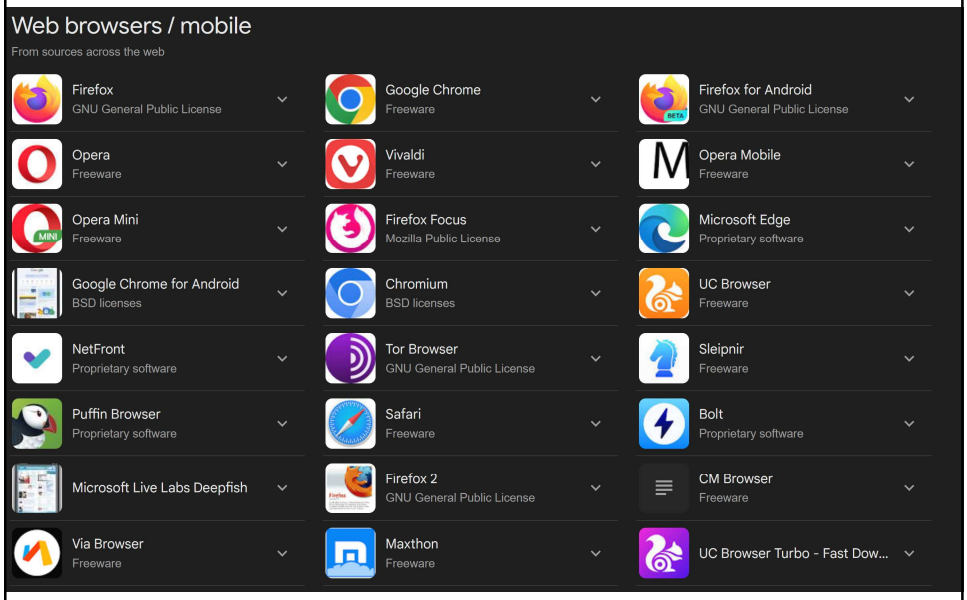

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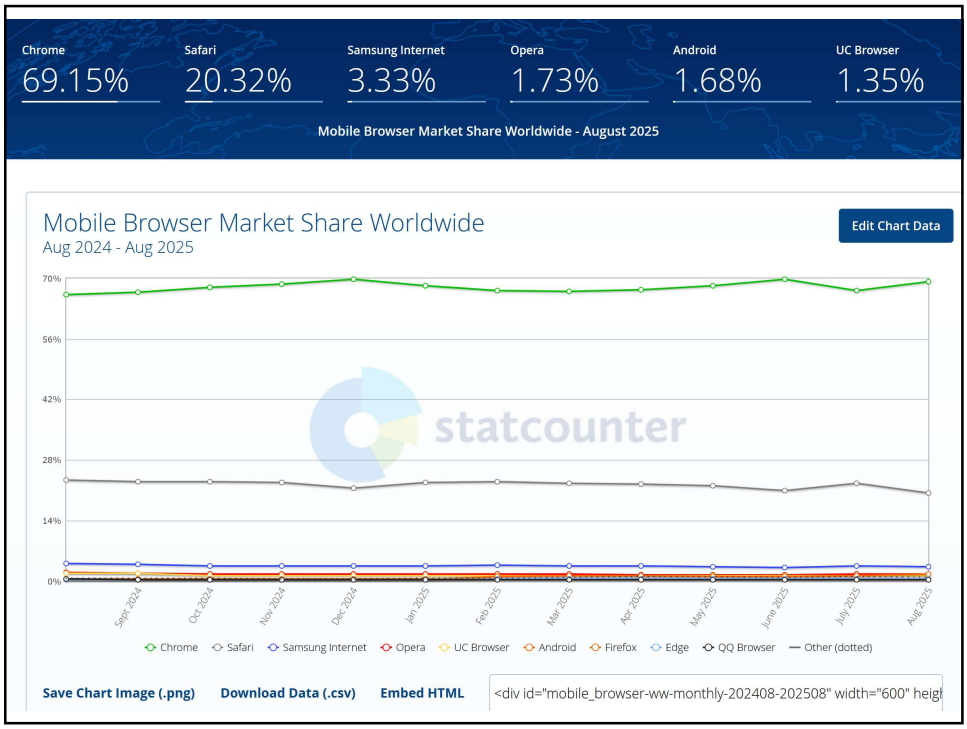

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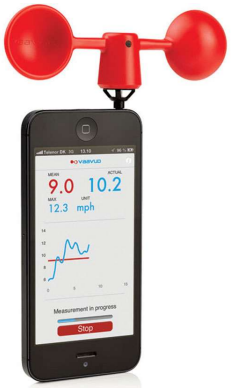
Applications (HW + SW)

- https://windandweathertools.com/the-best-smartphone-anemometers-wind-meters-with-iphone-android-apps/



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